

KANGERLUSSUAQ, GreenlandWMO: **042310**Lat: **67.017N**Lon: **50.717W**Elev: **53**StdP: **100.69**Time Zone: **-3.00 (GRL)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-36.9	-34.2	-41.0	0.1	-36.4	-38.2	0.1	-34.1	11.7	-2.4	10.0	-3.9	4.0	60	0.870

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	19.4	10.2	18.2	9.6	17.0	9.2	11.1	17.4	10.3	16.6	9.6	15.7	5.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.9	6.7	11.7	6.9	6.2	10.9	6.0	5.8	10.5	32.1	17.4	30.1	16.8	28.4	15.8	14.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.8	6.8	-38.4	21.6	3.7	0.9	-41.0	22.2	-43.2	22.8	-45.3	23.3	-48.0	23.9		
(5)			-38.5	12.5	3.7	0.8	-41.1	13.0	-43.3	13.5	-45.3	13.9	-48.0	14.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.2	-17.8	-19.1	-15.9	-5.6	3.8	10.0	11.1	8.9	3.7	-5.0	-10.8	-14.9
	DBStd	12.84	9.40	9.63	8.96	7.37	4.51	2.64	2.11	2.49	3.64	6.20	8.13	8.68
	HDD10.0	5293	863	816	803	469	194	32	13	50	192	466	624	771
	HDD18.3	8234	1122	1049	1062	719	449	249	226	292	440	724	874	1029
	CDD10.0	99	0	0	0	0	3	32	46	17	2	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.7	3.7	3.5	3.2	3.3	3.8	4.2	4.1	3.8	3.5	3.6	4.0	3.9
Precipitation	PrecAvg	176	8	6	4	10	12	13	26	33	22	16	13	10
	PrecMax	295	23	22	11	39	33	41	101	79	66	50	39	23
	PrecMin	104	0	1	0	1	0	0	1	7	5	3	1	1
	PrecStd	46	6	5	3	10	10	12	22	18	15	9	8	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	6.7	8.1	12.6	18.9	20.9	20.9	19.1	15.1	10.7	10.1	6.9
		MCWB	2.2	2.6	3.4	6.0	9.3	10.2	11.0	10.3	7.5	4.5	5.0	1.8
	2%	DB	3.0	3.2	4.0	8.6	15.3	19.2	19.7	17.6	12.8	7.8	6.0	4.1
		MCWB	-0.7	-0.8	0.6	3.3	7.3	9.9	10.5	9.7	6.4	3.4	2.0	0.5
	5%	DB	-0.2	0.3	0.3	6.4	13.0	18.0	18.6	16.2	10.8	5.4	3.5	1.2
		MCWB	-3.1	-3.0	-2.5	2.2	6.2	9.4	9.9	9.1	5.7	1.8	0.0	-1.6
	10%	DB	-4.5	-4.0	-2.6	4.1	10.9	16.3	17.2	14.7	9.2	3.0	1.1	-2.0
		MCWB	-6.4	-6.0	-4.9	0.9	5.1	8.6	9.4	8.5	5.0	0.3	-1.5	-4.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	3.0	3.6	6.3	9.6	12.0	12.5	12.0	8.6	5.4	5.2	2.3
		MCDB	6.2	5.9	7.8	11.9	17.1	18.0	19.1	17.2	13.2	9.2	9.4	6.0
	2%	WB	-0.4	-0.7	0.6	3.8	7.9	10.7	11.2	10.5	7.4	3.6	2.2	0.6
		MCDB	2.8	3.0	3.6	7.7	14.2	17.7	17.5	15.8	11.0	7.4	5.5	3.8
	5%	WB	-3.3	-3.0	-2.3	2.5	6.6	9.8	10.5	9.6	6.3	1.9	0.2	-1.6
		MCDB	-0.3	0.4	0.4	5.8	12.1	16.9	16.8	14.7	9.6	4.9	3.0	1.3
	10%	WB	-6.5	-6.0	-4.8	1.1	5.4	9.0	9.9	8.9	5.4	0.5	-1.5	-4.4
		MCDB	-4.5	-4.0	-2.5	3.9	10.1	15.5	16.2	13.8	8.5	2.6	1.3	-2.1
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.5	9.5	8.7	8.6	9.7	10.5	9.5	7.5	6.8	7.7	8.0
		MCDBR	12.7	13.1	11.8	10.1	12.0	12.8	13.1	12.3	9.5	8.7	10.6	10.6
		MCWBR	10.0	10.4	9.2	6.4	6.4	6.3	6.5	6.5	5.6	5.9	7.9	8.1
	5% WB	MCDBR	12.8	12.8	11.8	9.4	11.2	11.8	11.4	10.5	8.1	7.9	10.2	10.2
		MCWBR	10.2	10.3	9.2	6.2	6.4	6.2	6.3	6.0	5.2	5.5	7.8	8.0
Clear-Sky Solar Irradiance	taub	0.154	0.217	0.246	0.285	0.303	0.314	0.314	0.309	0.271	0.234	0.164	N/A	
	taud	1.919	2.161	2.277	2.250	2.394	2.483	2.534	2.528	2.532	2.260	1.935	N/A	
	Ebn at Noon	244	654	831	877	899	895	885	853	811	619	200	0	
	Edh at Noon	12	46	73	99	97	92	85	77	58	41	11	0	
All-Sky Solar Radiation	RadAvg	0.06	0.59	2.21	4.43	6.05	6.44	5.73	4.01	2.24	0.83	0.13	0.00	
	RadStd	0.00	0.06	0.12	0.28	0.38	0.49	0.51	0.25	0.17	0.08	0.01	0.00	

Historical Trends

	DBAvg	Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(46) Station Only	N/A	N/A	N/A	+0.78	N/A	-0.66	N/A	N/A	+22
(47) Regional (2 neighbors)	N/A	N/A	N/A	+1.04	N/A	N/A	N/A	N/A	+22

Nomenclature: See separate page